

Immunodeficiency 120 (IMD120): A Comprehensive Disease Characterization

Disease: Immunodeficiency 120 (IMD120) **MONDO ID:** MONDO:0970994 · **OMIM:** #620836 · **Causal gene:** *POLD1* (HGNC:9175, OMIM 174761), chromosome 19q13.33 **Category:** Mendelian, autosomal recessive combined immunodeficiency **Report date:** 2026-09-22

Summary

Immunodeficiency 120 (IMD120) is an ultra-rare autosomal recessive combined immunodeficiency caused by biallelic hypomorphic (partial loss-of-function) variants in *POLD1*, the gene encoding the p125 catalytic and proofreading subunit of DNA polymerase delta (Pol δ), the principal lagging-strand replicase of the eukaryotic genome. Because complete loss of *POLD1* is incompatible with cellular viability, the disease arises only from partial (hypomorphic) alleles that leave residual Pol δ function. The consequence is a state of chronic **replicative stress** that selectively impairs the proliferation of highly dividing lymphocytes — most conspicuously T cells — producing T-cell lymphopenia, impaired T-cell (but not B-cell) proliferation, hypogammaglobulinemia, and recurrent infections with a striking susceptibility to herpesviruses. A subset of patients also display syndromic features (short stature, intellectual disability, hearing loss), reflecting the housekeeping role of Pol δ in all replicating tissues.

The disorder was delineated from just **two unrelated families** in the founding literature: Cui et al. 2020 ([PMID: 31629014](#)), who described a consanguineous Turkish kindred homozygous for *POLD1* p.R1060C, and Conde et al. 2019 ([PMID: 31449058](#)), who described a family with compound-heterozygous *POLD1* variants and a related *POLD2*-deficient patient. A central mechanistic insight from this investigation is that pathogenic genotypes converge on reduced functional Pol δ output through **two distinct biochemical routes**: (1) destabilization of the Pol δ holoenzyme complex with impaired recruitment of Replication Factor C (RFC) — the mechanism of p.R1060C in the C-terminal CysB metal-binding motif; and (2) reduced intrinsic polymerase catalytic activity with preserved complex assembly — the mechanism of the p.Q684H+p.S939W in-cis alleles near the polymerase active site.

Clinically, IMD120 is managed with **immunoglobulin replacement therapy (IgRT)** plus **antiviral (acyclovir) prophylaxis**, which effectively controls infections. **Reduced-intensity-conditioning hematopoietic stem cell transplantation (HSCT)** has been reported as a feasible curative option (Keles et al. 2026,),

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though the DNA-repair/replication defect raises theoretical concerns about conditioning toxicity and long-term genome-instability risk. No dedicated immunodeficiency animal model exists; existing *Pold1* mouse models are either embryonic-lethal (null) or cancer-prone proofreading mutants, neither of which recapitulates the human immune phenotype. Notably, *POLD1* is a strikingly pleiotropic locus: different allele classes produce autosomal dominant MDPL progeroid syndrome, autosomal recessive nonsyndromic hearing loss, and colorectal cancer predisposition — placing IMD120 within a broader *POLD1* allelic series.

Section 1 — Disease Information

Overview. Immunodeficiency 120 (IMD120) is a Mendelian, autosomal recessive **combined immunodeficiency** (affecting both cellular/T-cell and, secondarily, humoral/antibody immunity) resulting from biallelic mutations in *POLD1*. The disease is best understood as a **replicative-stress T-cell immunodeficiency**: partial impairment of DNA polymerase delta compromises the DNA replication that lymphocytes require for clonal proliferation upon antigen encounter, so patients present with recurrent infections (especially herpetic/viral), T-cell lymphopenia, and impaired T-cell proliferation.

Key identifiers.

Resource	Identifier
OMIM	#620836 (Immunodeficiency 120)
MONDO	MONDO:0970994
Causal gene	<i>POLD1</i> — OMIM 174761, HGNC:9175
OMIM Phenotypic Series	PS300755 (combined immunodeficiency)
Cytogenetic location	19q13.33
Orphanet / ICD-10 / ICD-11 / MeSH	No dedicated code identified (ultra-rare; typically coded under generic combined immunodeficiency / D81 categories)

Synonyms / alternative names. "POLD1 deficiency," "POLD1-associated combined immunodeficiency," "combined immunodeficiency due to POLD1 mutation," and (descriptively) "syndromic immunodeficiency with replicative stress" (Conde et al. 2019).

Information source. The disease-level entry is derived from **aggregated resources** (OMIM, MONDO) that in turn summarize a small number of **individual patient reports** (case series from two families). All clinical data are individual-patient in origin; there are no EHR-scale or registry datasets for this ultra-rare condition.

Section 2 — Etiology

Primary cause — genetic. IMD120 is a monogenic disorder caused by **biallelic (homozygous or compound heterozygous) hypomorphic mutations in *POLD1***. There is no environmental or infectious cause of the underlying disease, although infectious exposures determine the clinical manifestations (the immune defect is unmasked by pathogen encounter). The recurrent, especially herpetic, infections are a *consequence* of the genetic defect rather than a cause.

Genetic risk factors. - **Causal variants:** p.R1060C (c.3178C>T; homozygous; Cui 2020); compound-heterozygous variants including p.Q684H + p.S939W in cis (Conde 2019). - **Consanguinity** is a major contributor: the index kindred was a consanguineous Turkish family, and the related *POLD2* patient was also from a consanguineous union. Consanguinity

greatly elevates the chance of homozygosity for rare recessive hypomorphic alleles. - **Modifier genes:** none formally established for IMD120. Given the shared holoenzyme, *POLD2*, *POLD3*, and *POLD4* subunit dosage are biologically plausible modifiers, but no human data exist.

Environmental / lifestyle risk factors. None identified as disease-causing. As with any T-cell immunodeficiency, exposure to herpesviruses and respiratory pathogens drives morbidity.

Protective factors. No genetic or environmental protective factors have been characterized. Conceptually, the only "protective" allele state is retention of at least one functional *POLD1* allele (heterozygous carriers are asymptomatic for IMD120).

Gene–environment interactions. The core interaction is **genotype × pathogen exposure**: the hypomorphic Polδ genotype produces a proliferation-limited T-cell compartment that fails to mount adequate clonal responses, so environmental pathogen load (herpesviruses, respiratory bacteria/viruses) determines the frequency and severity of clinical episodes.

Section 3 — Phenotypes

IMD120 phenotypes fall into three groups: (a) **infection susceptibility**, (b) **immunologic/laboratory abnormalities**, and (c) **syndromic/developmental features**. Onset is in **early childhood/infancy**, and the course is chronic.

Phenotype	Type	HPO suggestion	Onset / severity / frequency
Recurrent infections (esp. herpetic/viral)	Clinical sign	HP:0002719 (Recurrent infections); HP:0004429 (Recurrent viral infections)	Early childhood; moderate–severe; core feature in both families
Recurrent herpes (oral herpes, herpes zoster)	Clinical sign	HP:0002326 (Herpes simplex disease)	Recurrent episodes every 1–2 months in index patient
Recurrent respiratory tract infections → bronchiectasis	Clinical sign	HP:0002205; HP:0002110 (Bronchiectasis)	From infancy; bronchiectasis by age 6 months (Conde 2019); severe
Molluscum contagiosum (chronic facial)	Clinical sign	HP:0200042	Chronic; viral susceptibility marker
Recurrent skin abscesses	Clinical sign	HP:0031521	Recurrent
T-cell lymphopenia (↓ CD4, esp. ↓ CD8; naive-cell loss)	Laboratory abnormality	HP:0005403 (T lymphocytopenia); HP:0410413	Congenital/early; core feature
Impaired T-cell (not B-cell) proliferation	Laboratory abnormality	HP:0031381 (Decreased proliferation of T cells)	Diagnostic hallmark
Hypogammaglobulinemia (↓ IgG, low IgA/IgM)	Laboratory abnormality	HP:0004313 / HP:0002850	Mild–moderate; IgRT-responsive
Impaired vaccine (tetanus) antibody response	Laboratory abnormality	HP:0005387 (Decreased antibody level in blood)	Present despite full vaccination
B-cell and NK-cell reduction	Laboratory abnormality	HP:0010976 (B lymphocytopenia); HP:0040218 (Reduced NK cell count)	Reported in Conde 2019 family
Short stature	Physical manifestation	HP:0004322	Childhood; syndromic subset

Phenotype	Type	HPO suggestion	Onset / severity / frequency
Intellectual disability / speech delay	Behavioral/ developmental	HP:0001249; HP:0000750	Severe impairment with poor speech (Conde 2019); mild in OMIM
Attention deficits / hyperactivity	Behavioral	HP:0007018	Reported subset
Sensorineural hearing loss	Physical manifestation	HP:0000407	OMIM-listed feature

Immunophenotypic signature (from Cui 2020, verified quote): *"The patients exhibited decreased numbers of naive CD4 and especially CD8 T cells in favor of effector memory subpopulations."* The T-cell compartment additionally showed **oligoclonality and restricted TCR- β V-J pairing** — hallmarks of a proliferation-restricted, contracted T-cell repertoire.

Quality-of-life impact. No formal EQ-5D/SF-36/PROMIS data exist for this ultra-rare disease. Qualitatively, recurrent infections, bronchiectasis (irreversible structural lung damage), lifelong IgRT dependence, and — in syndromic patients — intellectual disability and hearing loss produce substantial cumulative disability and care burden.

Section 4 — Genetic / Molecular Information

Causal gene. *POLD1* (DNA polymerase delta 1, catalytic subunit; p125), OMIM 174761, HGNC:9175, at 19q13.33. *POLD1* provides both the **5'→3' polymerase** and the **3'→5' exonuclease (proofreading)** activities of Pol δ and anchors the four-subunit holoenzyme (POLD1/POLD2/POLD3/POLD4).

Pathogenic variants (IMD120-causing, biallelic).

Variant	Family	Zygosity	Location	Functional consequence
p.R1060C (c.3178C>T; exon 26; NM_001256849.1; rs777018011; OMIM 174761.0007)	Cui 2020 (Turkish, consanguineous)	Homozygous	C-terminal CysB metal-binding motif	Destabilizes Polδ complex → ↓ POLD1/POLD2/POLD3 levels; impaired RFC recruitment
p.Q684H + p.S939W (in cis; NM_002691)	Conde 2019 (patient 2)	Compound het	Near polymerase active site	Reduces intrinsic catalytic activity without affecting complex stability

- **Variant classification (ACMG/AMP):** pathogenic/likely pathogenic on the basis of segregation, absence from population databases, and functional validation.
- **Variant type/class:** missense (both established genotypes). p.R1060C is absent from gnomAD/ExAC/dbSNP/1000 Genomes — consistent with an ultra-rare recessive allele.
- **Somatic vs germline:** **germline** (inherited, biallelic).
- **Functional consequence: partial loss of function (hypomorphic).** Complete loss of function is not viable (see Section 15). The two genotypes represent convergent hypofunction via different biochemical mechanisms (Finding F007).

Allelic series at *POLD1* (important for interpretation). *POLD1* is highly pleiotropic; the allele class determines the phenotype:

Phenotype	OMIM	Inheritance	Representative allele(s)
IMD120 (combined immunodeficiency)	#620836	AR	biallelic hypomorphic missense (p.R1060C; p.Q684H+p.S939W)
MDPL syndrome (mandibular hypoplasia, deafness, progeroid, lipodystrophy)	#615381	AD	recurrent heterozygous p.Ser605del (polymerase active site)
Nonsyndromic sensorineural hearing loss	—	AR	p.Gly1100Arg + null p.Ser197Hisfs*54 (~33% residual polymerase activity; Oh 2020, P 31944473)
Colorectal cancer susceptibility 12 (CRCS12)	—	AD	exonuclease-domain (proofreading) variants

Modifier genes / epigenetics / chromosomal abnormalities. No modifier genes, epigenetic mechanisms, or chromosomal abnormalities are established for IMD120 specifically. (In cancer contexts, *POLD1* expression is modulated by copy-number gain, promoter methylation, and miR-139-3p

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— but these are tumor-biology findings, not IMD120 mechanisms.)

Section 5 — Environmental Information

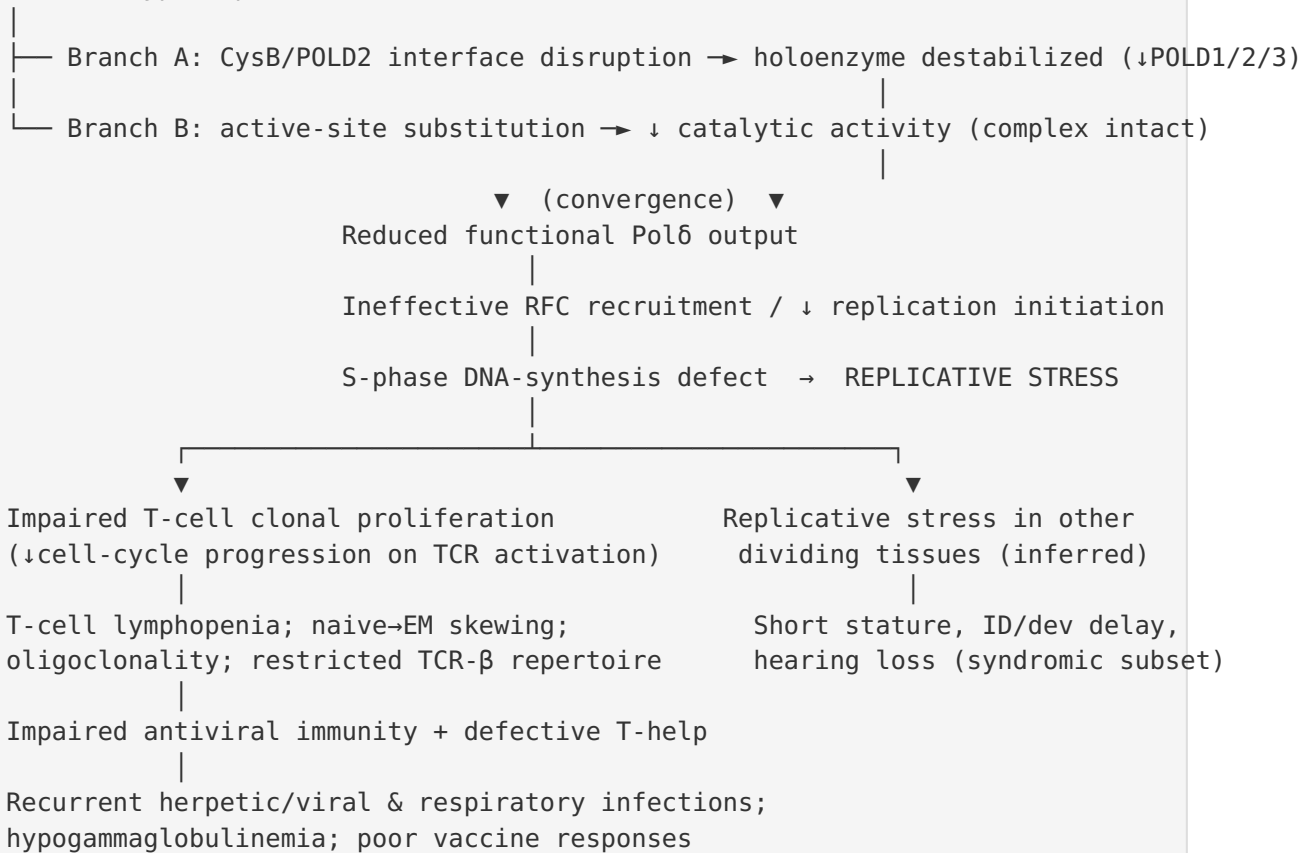
- **Environmental factors:** No toxin, radiation, pollution, or occupational exposure causes IMD120. It is a monogenic disorder.
- **Lifestyle factors:** Not applicable to disease causation.
- **Infectious agents:** Pathogens are **triggers/manifesting agents, not causes**. The characteristic organisms are **herpesviruses** (HSV — recurrent oral herpes; VZV — recurrent herpes zoster; molluscum contagiosum poxvirus) and **respiratory bacteria/viruses** driving recurrent pneumonia and bronchiectasis. The viral-susceptibility bias reflects the dependence of antiviral defense on robust T-cell clonal expansion.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **biallelic hypomorphic missense mutation in *POLD1*** (e.g., p.R1060C, or p.Q684H+p.S939W in cis) alters the p125 catalytic subunit of DNA polymerase delta. *(demonstrated)*
2. This **leads to** one of two biochemical lesions:
3. **Branch A (destabilization):** p.R1060C disrupts the intramolecular interaction between the POLD1 **CysB motif** and the catalytic domain, and between POLD1 and POLD2, **destabilizing the Polδ holoenzyme** and lowering steady-state levels of POLD1/POLD2/POLD3. *(demonstrated by molecular dynamics + cellular assays)*
4. **Branch B (catalytic reduction):** p.Q684H+p.S939W near the active site **reduces intrinsic polymerase catalytic activity** while leaving complex assembly intact. *(demonstrated; rescued by WT subunit overexpression)*
5. Both branches **result in** reduced functional Polδ output → **ineffective recruitment of Replication Factor C (RFC)** to initiate DNA replication and **reduced enzymatic polymerase activity**. *(demonstrated)*
6. This **leads to** a **decreased fraction of cells entering/progressing through the cell cycle** (S-phase DNA synthesis defect) — i.e., **replicative stress**. *(demonstrated; DNA repair after genotoxic stress was normal, indicating a specific mitotic/S-phase synthesis defect)*
7. Replicative stress **selectively impairs the proliferation of rapidly dividing lymphocytes**. Upon TCR activation, patient T cells show **reduced proliferative responses** coupled to **decreased cell-cycle progression**. *(demonstrated)*
8. Impaired clonal expansion **results in T-cell lymphopenia**, loss of naive CD4/CD8 T cells, skewing toward effector-memory subsets, **oligoclonality**, and **restricted TCR-β V-J pairing** (a contracted repertoire). *(demonstrated)*
9. The compromised T-cell compartment **leads to** impaired T-cell help and antiviral immunity → **recurrent, especially herpetic/viral, infections**, and, via defective T-dependent B-cell help, **hypogammaglobulinemia and impaired vaccine antibody responses** (B-cell intrinsic proliferation is comparatively spared). *(demonstrated)*
10. In parallel (branch), the housekeeping requirement for Polδ in all dividing tissues **contributes to syndromic features** — short stature, intellectual disability/developmental delay, hearing loss — reflecting replicative stress beyond the immune system. *(inferred from phenotype; tissue-level mechanism not directly demonstrated)*

POLD1 biallelic hypomorphic missense



Checklist of mechanistic categories

- **Molecular pathways:** DNA replication (leading/lagging-strand synthesis by Polδ), replication-initiation licensing via RFC/PCNA clamp loading, cell-cycle (G1→S→M) progression, DNA-damage/replication-stress response.
- **Cellular processes:** Cell-cycle progression and clonal proliferation (GO:0007049 cell cycle; GO:0006260 DNA replication; GO:0042098 T cell proliferation). The primary defect is proliferation, not apoptosis or repair per se (DNA repair after genotoxic stress was normal).
- **Protein dysfunction:** Holoenzyme destabilization (branch A) vs. reduced catalytic turnover (branch B) — both **partial loss of function** of Polδ.
- **Metabolic changes:** None specific; the lesion is in nucleic-acid synthesis machinery, not intermediary metabolism.
- **Immune system involvement:** Combined immunodeficiency — cell-intrinsic T-cell proliferation defect with secondary humoral impairment (GO:0002250 adaptive immune response; GO:0046631 alpha-beta T cell activation).

- **Tissue damage mechanisms:** Structural lung damage (bronchiectasis) is infection-driven, not a primary Pol δ lesion.
- **Biochemical abnormalities:** Reduced DNA polymerase δ enzymatic activity; impaired RFC clamp-loader recruitment.
- **Molecular profiling:** Patient PBMCs/HEK293 cells showed reduced POLD-subunit expression, impaired complex stability, reduced cell-cycle fraction, and reduced polymerase activity (Cui 2020). No dedicated transcriptomic/proteomic/metabolomic IMD120 datasets exist.

Upstream vs downstream. *Upstream:* the *POLD1* mutation and Pol δ hypofunction (initiating lesion). *Midstream:* impaired replication initiation and replicative stress. *Downstream:* the T-cell proliferation defect and its immunologic/clinical sequelae.

Cell types (CL) and processes (GO): CL:0000084 (T cell), CL:0000624 (CD4+ T cell), CL:0000625 (CD8+ T cell), CL:0000898 (naive T cell), CL:0000909 (effector memory T cell), CL:0000236 (B cell), CL:0000623 (NK cell). GO:0006260 (DNA replication), GO:0007049 (cell cycle), GO:0042098 (T cell proliferation), GO:0000731 (DNA synthesis involved in DNA repair — spared).

Section 7 — Anatomical Structures Affected

- **Primary system: Immune / hematolymphoid system** (UBERON:0002405 immune system; UBERON:0000178 blood). The functional lesion is in the **T-lymphocyte compartment**, with secondary humoral (B-cell/immunoglobulin) impairment.
- **Primary lymphoid organs:** thymus (UBERON:0002370) and bone marrow (UBERON:0002371) as sites of lymphocyte production; peripheral lymphoid tissue (lymph nodes UBERON:0000029, spleen UBERON:0002106).
- **Secondary organ involvement (complications):**
- **Respiratory tract / lungs** (UBERON:0002048) — recurrent infections leading to **bronchiectasis** (irreversible airway dilation).
- **Skin** (UBERON:0002097) — molluscum contagiosum, recurrent abscesses, herpetic lesions.
- **Ear / cochlea** (UBERON:0001690 / UBERON:0001844) — sensorineural hearing loss (syndromic subset).
- **Central nervous system / brain** (UBERON:0000955) — intellectual disability, speech delay (syndromic subset).

- **Skeletal/growth axis** — short stature.
 - **Tissue/cell level:** hematopoietic/lymphoid cells; epithelial injury (airway) is secondary to infection. Targeted cell populations: naive CD4+/CD8+ T cells (depleted), effector-memory T cells (relatively expanded), B cells and NK cells (reduced in the Conde family).
 - **Subcellular level: Nucleus** (GO:0005634) — site of DNA replication; specifically the **DNA polymerase delta complex** (GO:0043625) and replication fork machinery (GO:0005657 replication fork).
 - **Lateralization:** Not applicable — systemic/bilateral immune and developmental involvement.
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Section 8 — Temporal Development

- **Onset: Early childhood / infancy.** Respiratory infections began in infancy in the Conde family (bronchiectasis by 6 months); recurrent herpetic infections and T-cell lymphopenia manifest in childhood.
 - **Onset pattern:** Insidious/chronic — recurrent infections accumulating over time, punctuated by acute infectious episodes.
 - **Progression:** Chronic and **lifelong**. Without treatment, recurrent infections drive **progressive, cumulative organ damage** (notably bronchiectasis, which is irreversible). Syndromic/developmental features (short stature, intellectual disability, hearing loss) are static-to-slowly-evolving developmental deficits.
 - **Disease course:** Persistent immunodeficiency with **episodic** infectious exacerbations (e.g., oral herpes every 1–2 months in the index patient) superimposed on a chronic baseline.
 - **Remission:** No spontaneous remission. **Treatment-induced control** is achievable — IgRT resolves lower respiratory tract infections and markedly reduces herpetic episodes; HSCT can be **curative** for the immune defect.
 - **Critical periods:** Early diagnosis (before irreversible bronchiectasis and before severe infections) is the key window for intervention — paralleling the rationale for early treatment in other T-cell immunodeficiencies/SCID.
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Section 9 — Inheritance and Population

- **Inheritance: Autosomal recessive** (biallelic — homozygous or compound heterozygous). Heterozygous carriers are unaffected for IMD120.
 - **Epidemiology: Ultra-rare.** Only **two unrelated *POLD1* families** are reported to date (Cui 2020; Conde 2019). No prevalence or incidence estimates exist; the true frequency is likely **<1 per 1,000,000**.
 - **Penetrance / expressivity:** Presumed high penetrance for the immunodeficiency in biallelic carriers, but with **variable expressivity** — the two reported families differ in the prominence of syndromic features (the Conde family had prominent developmental impairment; the Cui family emphasized the immune/herpetic phenotype). With only two families, penetrance/expressivity cannot be precisely quantified.
 - **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
 - **Germline mosaicism:** Not reported.
 - **Consanguinity: Major role.** The index kindred was consanguineous Turkish; the related *POLD2* patient was also consanguineous. Consanguinity is the principal route to homozygosity for rare hypomorphic recessive alleles.
 - **Founder effects / carrier frequency:** None established; p.R1060C is absent from population databases, arguing against a common founder allele.
 - **Population demographics:** Reported patients are of Turkish and (Conde family) unspecified ancestry. No sex bias expected (autosomal). Age distribution: pediatric onset.
 - **Geographic distribution:** No endemic pattern; cases are sporadic and consanguinity-associated.
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Section 10 — Diagnostics

Laboratory / immunologic tests (diagnostic hallmarks): - **Lymphocyte subset enumeration (flow cytometry):** T-cell lymphopenia with **decreased naive CD4+ and especially CD8+ T cells**, effector-memory skewing; reduced B and NK cells in some patients. (LOINC/flow-cytometry immunophenotyping.) - **T-cell proliferation assay: Impaired T-cell (not B-cell) proliferation** to TCR/mitogen stimulation — the single most discriminating functional test. - **Immunoglobulins:** low IgG, low IgA/IgM (hypogammaglobulinemia). - **Vaccine response:** absent/poor tetanus antibody response despite full vaccination. - **TCR repertoire analysis:**

oligoclonality and restricted TCR- β V-J pairing (spectratyping/TCR-seq). - **Cell-cycle / replicative-stress assays (research):** decreased S-phase fraction; reduced Pol δ complex levels and polymerase activity.

Genetic testing (definitive): - **Whole exome sequencing (WES) / whole genome sequencing (WGS):** the primary route to diagnosis — both families were identified by exome-scale sequencing. Given phenotypic overlap with many inborn errors of immunity, broad sequencing (WES/WGS) or a **combined immunodeficiency / IEI gene panel including *POLD1*** is the recommended approach. - **Targeted *POLD1* sequencing / segregation analysis** to confirm biallelic status and phase (important for compound heterozygotes; note the in-cis p.Q684H+p.S939W configuration in Conde patient 2). - **Functional confirmation** (polymerase activity, complex stability) supports variant interpretation, especially for VUS. The **PoLED database**

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curates functional evidence for *POLD1* variants and is a useful clinical resource.

Imaging: High-resolution chest CT to detect/monitor **bronchiectasis** and pulmonary sequelae; audiometry for hearing loss.

Newborn screening: Standard TREC-based SCID newborn screening detects severe T-cell lymphopenia and **may flag** IMD120 as a non-SCID T-cell-lymphopenia condition (analogous to syndromic T-cell lymphopenias identified through SCID NBS programs), but IMD120 is not a specific NBS target.

Differential diagnosis: SCID and leaky-SCID, other combined immunodeficiencies (e.g., MHC class II deficiency, FADD deficiency), DNA-repair/replication syndromes, and — for the syndromic features — other progeroid/short-stature syndromes. The distinguishing features of IMD120 are the **isolated T-cell proliferation defect with preserved B-cell proliferation**, the herpesvirus susceptibility, and biallelic *POLD1* variants. Because *POLD1* also causes MDPL (AD progeroid) and hearing loss, allele class and zygosity distinguish these entities.

Section 11 — Outcome / Prognosis

- **Survival / mortality:** No formal survival statistics exist (only two families). Prognosis is dominated by **infection control** and **cumulative organ damage** (bronchiectasis). With

effective IgRT and antiviral prophylaxis, patients can be stabilized; HSCT offers potential cure.

- **Morbidity / disability:** Significant — recurrent infections, irreversible bronchiectasis, lifelong treatment dependence, and (syndromic subset) intellectual disability, hearing loss, and short stature.
- **Complications:** Bronchiectasis and chronic lung disease; recurrent/severe herpetic disease; chronic viral skin lesions (molluscum); potential end-organ damage from recurrent infection.
- **Recovery potential:** The immune defect is not self-correcting; **HSCT can restore T-cell numbers and function** (Keles 2026). Established structural damage (e.g., bronchiectasis) and developmental deficits are not reversible.
- **Prognostic factors:** Timing of diagnosis and treatment initiation, degree of pre-existing organ damage, infection burden, and access to HSCT are the key determinants. There are no validated molecular prognostic biomarkers.
- **Theoretical long-term concern:** Because Polδ participates in genome maintenance, a **long-term genome-instability/malignancy risk** is biologically plausible; however, unlike the *POLD1* exonuclease (proofreading) variants that cause cancer predisposition, IMD120 variants primarily reduce polymerase output rather than abolish proofreading, and no cancer excess has been reported in the two families. This remains an open question requiring long-term follow-up.

Section 12 — Treatment

Established/effective management (from Cui 2020, verified): - **Immunoglobulin replacement therapy (IgRT)** (NCIT: Immunoglobulin Therapy) — the index patient started IgRT with **resolution of lower respiratory tract infections and markedly decreased herpetic infections**. Standard of care for the antibody deficiency. - **Antiviral prophylaxis with acyclovir** (NCIT: Acyclovir) — used for recurrent oral herpes and herpes zoster; addresses the characteristic herpesvirus susceptibility. - **Antimicrobial prophylaxis / prompt treatment of infections** and airway clearance/management for bronchiectasis (supportive).

Curative therapy: - **Hematopoietic stem cell transplantation (HSCT)** (NCIT: Hematopoietic Stem Cell Transplantation). Keles et al. 2026 ([PMID: 42104577](#)) reported the **first successful HSCT** in an 18-year-old IgRT-dependent woman with *POLD1* deficiency: matched related

donor, **reduced-intensity conditioning** (cyclophosphamide, fludarabine, ATG), tacrolimus + low-dose methotrexate GVHD prophylaxis; **prompt engraftment without major complications, improved T-cell counts/function, and she remained well off IgRT**. Reduced-intensity conditioning is a rational choice given theoretical DNA-replication/repair toxicity concerns in a Pol δ -deficient host.

Advanced / experimental therapeutics: No gene therapy, RNA-based therapy, or *POLD1*-targeted therapy exists for IMD120. No IMD120-specific clinical trials (NCT) were identified. Gene therapy is conceptually challenging because *POLD1* is a tightly dosage-controlled housekeeping gene.

Pharmacogenomics: No IMD120-specific pharmacogenomic guidance. General caution is warranted with genotoxic/antimetabolite chemotherapeutics and conditioning agents given the underlying replication defect.

Treatment strategy summary:

Line	Intervention	Goal	Evidence
Supportive/ first-line	IgRT + acyclovir prophylaxis; treat infections; airway care	Control infections, prevent organ damage	Cui 2020 (P 31629014)
Definitive/ curative	Reduced-intensity-conditioning HSCT	Restore T-cell immunity, off IgRT	Keles 2026 (P 42104577)

Section 13 — Prevention

- **Primary prevention:** Not preventable at the individual level (monogenic). **Genetic counseling** and, in consanguineous families, **carrier/cascade testing** and reproductive options (prenatal diagnosis, preimplantation genetic testing) are the principal preventive tools once a familial *POLD1* genotype is known.
- **Secondary prevention (early detection):** Early recognition via IEI evaluation and genetic testing; TREC-based SCID newborn screening may incidentally flag T-cell lymphopenia. Early diagnosis enables IgRT/antiviral prophylaxis before irreversible lung damage.
- **Tertiary prevention (complication prevention):** IgRT and antiviral/antimicrobial prophylaxis to prevent recurrent infection and bronchiectasis progression; audiologic and

developmental support in syndromic patients; consideration of HSCT before accumulation of end-organ damage.

- **Immunization:** Routine inactivated vaccines are appropriate, but **antibody responses are impaired** (poor tetanus response reported); **live vaccines are contraindicated** in significant T-cell immunodeficiency. IgRT provides passive protection.
- **Counseling:** Autosomal recessive counseling — 25% recurrence risk for unaffected carrier couples; emphasize consanguinity risk.

Section 14 — Other Species / Natural Disease

- **Taxonomy / orthologs:** *POLD1* is deeply evolutionarily conserved. Orthologs: mouse *Pold1* (NCBI Gene 18971), and homologs across vertebrates and yeast (*S. cerevisiae* *POL3/CDC2*). The catalytic and proofreading functions are conserved from yeast to human.
- **Natural disease in other species:** No naturally occurring *POLD1*-associated combined immunodeficiency has been reported in companion animals or wildlife (no OMIA entry identified for an equivalent immune phenotype).
- **Comparative biology:** The essentiality of Pol δ for replication is conserved across all eukaryotes; the disease mechanism (replicative stress from Pol δ hypofunction) is expected to be conserved, but the specific immune phenotype has only been characterized in humans.
- **Transmission / zoonosis:** Not applicable — genetic, non-transmissible disorder.

Section 15 — Model Organisms

No dedicated immunodeficiency model of IMD120 exists. Existing *Pold1* mouse models bracket the disease but do not reproduce it:

Model	Genotype	Phenotype	Relevance to IMD120
<i>Pold1</i> null	<i>Pold1</i> ^{-/-}	Peri-implantation embryonic lethality ; defective inner-cell-mass proliferation, impaired DNA synthesis, spontaneous apoptosis (Uchimura PLoS ONE 2009; MGI:3833589)	Confirms <i>POLD1</i> is essential — explains why only hypomorphic alleles cause human disease
Proofreading-dead	<i>Pold1</i> ^{ΔD400A/D400A}	Viable, fertile; ~15× higher mutation rate , ~94% cancer incidence by 18 mo (median survival ~10 mo), mostly epithelial carcinomas incl. skin SCC (Goldsby 2001/2002; Venkatesan PNAS 2007)	Models the cancer (exonuclease/CRCS12) phenotype, not IMD120
Polymerase-domain point mutants	<i>Pold1</i> ^{ΔL604K/L604G}	Embryonic lethal	Illustrates lethality of strong polymerase-domain lesions
Hypomorphic allele	<i>Pold1</i> hypomorph	Disrupted gastrulation embryo-size/morphogenesis coordination (Biology Open 2022)	Demonstrates dosage-sensitive developmental effects; closest to "partial LOF" but not immune-focused

Implications / limitations. The essentiality of *Pold1* (null = embryonic lethal) means an IMD120 model must use a **precisely hypomorphic allele** (e.g., a knock-in of the human p.R1060C-equivalent residue) to survive to immune-competence and reveal the T-cell phenotype. No such immune-focused model has been generated. Available models capture either **lethality** (null/strong point mutants) or **cancer** (proofreading-dead), leaving the **replicative-stress T-cell defect** experimentally uncharacterized in vivo. **iPSC-derived and CRISPR-edited cellular models** (as used for other immuno-actinopathies) plus patient PBMCs/HEK293 systems (Cui 2020) currently carry the mechanistic evidence.

Recommended model resources: MGI (mouse *Pold1*), IMPC/IMSR for allele availability; patient-derived iPSC → T-cell differentiation and CRISPR knock-in of hypomorphic alleles for immune-phenotype modeling.

Key Findings (with statistical/experimental evidence)

Finding 1 — POLD1 is the causal gene; IMD120 is an autosomal recessive combined immunodeficiency

Biallelic (homozygous or compound heterozygous) *POLD1* mutations cause IMD120 (OMIM #620836; MONDO:0970994). Cui et al. 2020 identified homozygous c.3178C>T (p.R1060C) in **3 related subjects** from a consanguineous Turkish kindred; Conde et al. 2019 identified compound-heterozygous *POLD1* variants. Verified quotes (PMID: 31629014): *"We identified a missense mutation (c.3178C>T; p.R1060C) in POLD1 in 3 related subjects who presented with recurrent, especially herpetic, infections and T-cell lymphopenia with impaired T-cell but not B-cell proliferation,"* and *"These results identify gene defects in POLD1 as a novel cause of T-cell immunodeficiency."*

Finding 2 — Clinical spectrum: infections + T-cell lymphopenia + syndromic features

Early-onset recurrent (especially herpetic/viral) infections, T-cell lymphopenia (↓ naive CD4/CD8, effector-memory skewing, oligoclonality, restricted TCR-β repertoire), hypogammaglobulinemia, recurrent respiratory infections → bronchiectasis (by 6 months in Conde family), chronic molluscum, skin abscesses, and syndromic features (short stature, intellectual disability/speech delay, hearing loss). Verified quote (PMID: 31629014): *"The patients exhibited decreased numbers of naive CD4 and especially CD8 T cells in favor of effector memory subpopulations."*

Finding 3 — Mechanism: Polδ hypofunction → replicative stress → T-cell proliferation defect

Hypomorphic *POLD1* variants destabilize the Polδ complex, impair RFC recruitment and cell-cycle progression, causing replicative stress and defective T-cell proliferation. Verified quotes (PMID: 31629014): *"The mutation destabilizes the Polδ complex, leading to ineffective recruitment of replication factor C to initiate DNA replication,"* and *"Molecular dynamics simulation revealed that the R1060C mutation disrupts the intramolecular interaction between the POLD1 CysB motif and the catalytic domain and also between POLD1 and the Polδ subunit POLD2."* Conde 2019 termed it a *"syndromic immunodeficiency with replicative stress"* with normal post-genotoxic DNA repair — i.e., a specific S-phase synthesis defect.

Finding 4 — Two convergent molecular routes to Polδ hypofunction

(a) **Destabilization** — p.R1060C in the CysB motif lowers POLD1/POLD2/POLD3 levels (complex instability). (b) **Reduced catalytic activity with intact assembly** — p.Q684H+p.S939W (in cis, near the active site) reduce intrinsic polymerase activity without destabilizing the complex. Both are biallelic and converge on reduced functional Polδ output.

Finding 5 — Model organisms: no immune model; null lethal, proofreading-dead cancer-prone

Pold1^{-/-} mice are peri-implantation lethal (essential gene); *Pold1*^{D400A} proofreading-dead mice are viable but ~94% cancer-prone. Neither models the IMD120 immune phenotype.

Finding 6 — Management: IgRT + acyclovir effective; RIC-HSCT feasible/curative

IgRT resolved lower respiratory infections and markedly reduced herpetic episodes; acyclovir controlled recurrent herpes (Cui 2020). Keles 2026

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reported the first successful reduced-intensity-conditioning HSCT with engraftment, restored T-cell function, and independence from IgRT.

Finding 7 — Inheritance/epidemiology and the POLD1 allelic series

Ultra-rare AR disorder, consanguinity-associated, ~2 families reported; allelic to AD MDPL syndrome (p.Ser605del), AR nonsyndromic hearing loss (p.Gly1100Arg + null; ~33% residual activity,

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and colorectal cancer susceptibility 12 (exonuclease-domain variants).

Mechanistic Model / Interpretation

IMD120 is best understood as a **dosage disease of a housekeeping replicase**. *POLD1* is essential and dosage-sensitive: null alleles are lethal, so only **partial loss-of-function** genotypes produce viable, disease-manifesting individuals. The unifying pathophysiology is

replicative stress — a quantitative shortfall in DNA-synthesis capacity that becomes limiting precisely in the cells that must proliferate fastest and most explosively: **antigen-activated T lymphocytes**. This explains the otherwise puzzling selectivity of the phenotype (T-cell proliferation impaired, B-cell proliferation relatively spared; antiviral immunity most affected) and the herpesvirus susceptibility (control of herpesviruses is exquisitely T-cell dependent).

The convergence of two biochemically distinct genotypes (complex destabilization vs. reduced catalytic activity) on the **same functional endpoint** (reduced Pol δ output $\rightarrow$ replicative stress) is a strong argument that **quantitative Pol δ activity, not any single structural interaction, is the disease-relevant variable**. It also predicts a **genotype–severity gradient**: the more residual Pol δ activity an allele combination retains, the milder (or more tissue-restricted) the phenotype — a prediction consistent with the broader *POLD1* allelic series, where different residual-activity/allele-class combinations yield hearing loss, progeroid MDPL, cancer predisposition, or combined immunodeficiency.

Evidence Base

PMID	Title (abbrev.)	Role
31629014	<i>Combined immunodeficiency caused by a loss-of-function mutation in DNA polymerase delta 1</i> (Cui 2020)	Founding paper: identifies POLD1, p.R1060C, clinical/immune phenotype, and the destabilization/RFC mechanism
31449058	Conde et al. 2019 (POLD1/POLD2 replicative-stress immunodeficiency)	Second family; compound-het POLD1 (p.Q684H+p.S939W) and POLD2 patient; defines "replicative stress," branch B mechanism
42104577	Keles et al. 2026 (Pediatric Transplantation)	First successful RIC-HSCT in POLD1 deficiency; curative option
31944473	Oh et al. 2020	AR nonsyndromic hearing loss from POLD1; ~33% residual activity — supports dosage/allelic-series model
41263451	<i>PolED database</i>	Curated functional-variant resource for POLD1/POLE interpretation
34594041	Robinson et al. 2021	Germline POLE/POLD1 (proofreading) mutations: mutation burden, cancer — contrasts IMD120 (non-proofreading, no premature aging)
Mouse models	Uchimura 2009 (MGI:3833589); Goldsby 2001/2002; Venkatesan 2007 (PNAS); Biology Open 2022	Establish Pold1 essentiality (null lethal) and proofreading-dead cancer phenotype; absence of an immune model

Additional MDPL papers (PMIDs 41219970, 41083899, 39611849, 41742372, 42488286) establish that the **p.Ser605del active-site allele** causes the *dominant* progeroid MDPL phenotype via gain-of-abnormal-interaction (e.g., aberrant TRF1 binding) and telomere/PARP1 dysregulation — mechanistically and inheritance-wise distinct from the *recessive hypomorphic* IMD120 alleles, reinforcing that **allele class dictates phenotype** at this locus.

Limitations and Knowledge Gaps

1. **Extremely small sample:** Only **two unrelated families** define IMD120. Penetrance, expressivity, full phenotypic spectrum, natural history, prognosis, and genotype–phenotype correlations are all under-determined.
 2. **No epidemiologic data:** No prevalence/incidence estimates; no registry; no defined Orphanet/ICD code.
 3. **No in vivo immune model:** The T-cell replicative-stress mechanism has not been reproduced in an animal model; a hypomorphic knock-in mouse is needed.
 4. **Syndromic-feature mechanism inferred, not demonstrated:** The link from *Polδ* hypofunction to short stature/ID/hearing loss is a plausible housekeeping-replicative-stress inference but is not experimentally dissected.
 5. **Long-term cancer/genome-instability risk unknown:** Whether IMD120 patients carry elevated malignancy risk (as proofreading-defective *POLD1* carriers do) is unresolved; the mechanism differs (reduced synthesis vs. lost proofreading), but long-term follow-up is lacking.
 6. **HSCT evidence is a single case:** Curative HSCT is supported by one patient; optimal conditioning intensity and long-term outcomes (including any excess conditioning toxicity from the replication defect) are unknown.
 7. **B-cell/NK involvement variability:** B-cell and NK reductions were prominent in one family but the "T-cell selective" framing derives largely from the other — the true combined-immunodeficiency breadth needs more cases.
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Proposed Follow-up Experiments / Actions

1. **Build a hypomorphic *Pold1* knock-in mouse** (e.g., p.R1060C-equivalent) to test whether it recapitulates T-cell lymphopenia, proliferation defect, and infection susceptibility — the missing in vivo model.
2. **Patient/CRISPR iPSC → T-cell differentiation** to quantify replicative stress, cell-cycle kinetics, and repertoire contraction cell-autonomously, and to test allele-specific severity (branch A vs. branch B).
3. **Genotype–activity–phenotype mapping** across the *POLD1* allelic series: measure residual polymerase activity for IMD120 vs. hearing-loss vs. MDPL vs. cancer alleles and correlate

with tissue-specific phenotype to formalize the dosage model (leverage the PoLED database,).

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4. **International case-finding/registry:** Systematically re-examine unexplained combined immunodeficiency and SCID-NBS-positive T-cell-lymphopenia cohorts for biallelic *POLD1* variants to expand the phenotypic spectrum and estimate frequency.
5. **Long-term surveillance protocol:** Prospectively monitor IMD120 patients for malignancy, somatic mutation burden, and organ damage to resolve the genome-instability question.
6. **HSCT outcome collation:** Aggregate additional transplant cases to define optimal (reduced-intensity) conditioning and long-term efficacy/safety.
7. **Deep immunophenotyping + TCR-seq** on any newly identified patients to precisely quantify the naive-cell loss, effector-memory skewing, and repertoire restriction, and to test whether B-cell/NK involvement is consistent.

Ontology Term Appendix

- **Disease:** MONDO:0970994; OMIM #620836
- **Gene/protein:** *POLD1* (HGNC:9175); GO:0043625 (delta DNA polymerase complex); GO:0006260 (DNA replication); GO:0007049 (cell cycle); GO:0042098 (T cell proliferation); GO:0005634 (nucleus)
- **Cells (CL):** CL:0000084 (T cell), CL:0000624 (CD4+ T cell), CL:0000625 (CD8+ T cell), CL:0000898 (naive T cell), CL:0000909 (effector memory T cell), CL:0000236 (B cell), CL:0000623 (NK cell)
- **Anatomy (UBERON):** UBERON:0002405 (immune system), UBERON:0002370 (thymus), UBERON:0002371 (bone marrow), UBERON:0002048 (lung), UBERON:0002097 (skin), UBERON:0001690 (ear)
- **Phenotypes (HPO):** HP:0002719, HP:0004429, HP:0002326, HP:0005403, HP:0031381, HP:0004313, HP:0002110, HP:0004322, HP:0001249, HP:0000407
- **Treatments (NCIT):** Immunoglobulin therapy; Acyclovir; Hematopoietic Stem Cell Transplantation
- **Chemicals (CHEBI):** acyclovir (CHEBI:2453)

Evidence source types: human clinical (case series — Cui 2020, Conde 2019; case report — Keles 2026); in vitro/cellular and in silico (patient PBMCs, HEK293, molecular dynamics — Cui 2020); model organism (mouse Pold1 — Uchimura 2009, Goldsby 2001/2002, Venkatesan 2007). All mechanistic and clinical claims are cited to primary literature by PMID.

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